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Studierichting: Informatica
Oefeningenreeks 3A nr. 2.6

Bereken de afgeleide van $f(x) = -3x^2$ voor $a \in \{-2, 1, 5\}$

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

- $a = -2$

$$f'(-2) = \lim_{h \to 0} \frac{-3(h-2)^2 - f(-2)}{h}$$

$$= \lim_{h \to 0} \frac{-3(h^2 - 4h + 4) + 12}{h}$$

$$= \lim_{h \to 0} \frac{-3h^2 + 12h}{h}$$

$$= \lim_{h \to 0} (-3h + 12) = 12$$

- $a = 1$

$$f'(1) = \lim_{h \to 0} \frac{-3(h+1)^2 - f(1)}{h}$$

$$= \lim_{h \to 0} \frac{-3(h^2 + 2h + 1) - 3}{h}$$

$$= \lim_{h \to 0} \frac{-3h^2 - 6h}{h}$$

$$= \lim_{h \to 0} (-3h - 6) = -6$$

- $a = 5$

$$f'(5) = \lim_{h \to 0} \frac{-3(h+5)^2 - f(5)}{h}$$

$$= \lim_{h \to 0} \frac{-3(h^2 + 10h + 25) - 75}{h}$$

$$= \lim_{h \to 0} \frac{-3h^2 - 30h}{h}$$

$$= \lim_{h \to 0} (-3h - 30) = -30$$

References